INTERNSHIP REPORT

AI PRODUCT ENGINEERING INTERNSHIP AT PHYSICS WALLAH (PW) - PWIOI TEAM

SUBMITTED TO:ABHISHEK GUPTA

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PHYSICSWALLAH (PW)

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ABSTRACT

This report presents an in-depth analysis of the EdTech Industry, specifically focusing on PhysicsWallah (PW), one of India's leading education technology companies. Conducted during an internship as a AI Product Engineer under the PWIOI team, this study aims to provide a comprehensive overview of PW's product operations, my responsibilities as an intern, and the suggestions formulated based on observations and research.

The report begins with an introduction to the EdTech Industry in India, detailing its structure, major players, and current trends. This is followed by an overview of PhysicsWallah as an organization - its history, growth, vision, and key programs. Special emphasis is placed on the PWIOI (PhysicsWallah Institute of Innovation) initiative, its strategic importance, and operational framework.

The core of the report describes my internship experience, covering day-to-day responsibilities including dashboard development on Looker Studio, workflow automation using Python, Telegram bot building, Mettle video extraction automation, GCMS website management, and BRD documentation. Each deliverable is detailed with context, methodology, and outcomes.

My internship duties included building end-to-end Business Intelligence dashboards, creating Python automation scripts to eliminate manual data work, managing daily walk-in reports, and contributing to product documentation through BRDs. The insights gained from these activities are presented, along with specific suggestions to improve operational efficiency.

Key findings suggest that while PW's PWIOI team has strong operational foundations, there are significant opportunities for further automation, data centralization, and dashboard standardization. The report concludes with recommendations that address these areas, aiming to support PW in maintaining its operational edge and achieving data-driven growth.

This report serves as a reflection of my internship experience and a strategic document that could potentially benefit PhysicsWallah's PWIOI Product team. The analysis and recommendations provided herein are based on empirical data, team interactions, and personal observations made during the internship period.

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EXECUTIVE SUMMARY

INTRODUCTION TO THE EDTECH INDUSTRY

The EdTech Industry is one of India's fastest-growing sectors, driven by digital penetration, affordable smartphones, and a large student population. This report begins with an overview of the EdTech ecosystem, highlighting its significance, key players, market dynamics, and the regulatory environment.

OVERVIEW OF PHYSICSWALLAH (PW)

A detailed overview of PhysicsWallah is presented, covering its history, organizational structure, key programs (PWIOI, Vidyapeeth, Alakh IIT), and strategic importance within India's education sector. Special emphasis is placed on the PWIOI program - its structure, operations, and strategic mission of creating a world-class institute for innovation.

INTRODUCTION TO THE PWIOI TEAM AND PRODUCT ENGINEERING

The report delves into the PWIOI operational structure, including dashboard management, automation pipelines, Telegram marketing, and website management. The role of the AI Product Engineer intern is explained, including key responsibility areas and work systems.

INTERNSHIP DUTIES AND RESPONSIBILITIES

The internship involved building Looker Studio dashboards, Python automation scripts, Telegram bots, video extraction tools, and contributing to BRDs. The report outlines these responsibilities in detail, providing a week-wise account of all work conducted.

KEY PROJECTS AND DELIVERABLES

Major projects are discussed: PWIOI Admission and Sales Dashboards, Mettle Video Downloader, Telegram Bot, multiple Excel/Sheet automations, GCMS website updates, BTL/Mitra tracking dashboards, and BRD documentation. Methodologies, outcomes, and live links are provided.

ANALYSIS AND RECOMMENDATIONS

The report provides actionable recommendations for further automation, dashboard standardization, and data centralization. These suggestions aim to improve efficiency, reduce manual workload, and enhance decision-making speed for the PWIOI leadership team.

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1. INTRODUCTION TO THE EDTECH INDUSTRY

1.1 OVERVIEW OF THE EDTECH INDUSTRY

The Education Technology (EdTech) industry is one of the most transformative sectors in the global knowledge economy. EdTech companies build platforms, tools, and content that enable learning at scale - bridging the gap between quality education and students who lack access to traditional institutions. These companies leverage digital infrastructure including mobile apps, video streaming, AI-powered adaptive learning, and live interactive sessions to deliver education at a fraction of the cost of traditional schooling.

In India, the EdTech revolution has been particularly profound. With over 250 million students in K-12 and a large aspirational base seeking competitive exam preparation (JEE, NEET, UPSC), India has become one of the world's largest EdTech markets. PhysicsWallah (PW) stands at the forefront of this movement - democratizing quality education by offering affordable and highly effective learning solutions.

1.2 SIGNIFICANCE OF THE EDTECH INDUSTRY

The EdTech industry is integral to India's human capital development. Its significance can be understood through several key dimensions:

• Access to Quality Education: EdTech platforms deliver IIT/NEET-level faculty to students in Tier 2 and Tier 3 cities at affordable prices.

• Economic Impact: India's EdTech sector is valued at over $5 billion and is expected to reach $30 billion by 2030, contributing to employment and the digital economy.

• Technological Innovation: AI-based personalized learning, video-based content, live doubt sessions, and assessment platforms are reshaping how students learn.

• Scalability: Unlike physical classrooms, EdTech platforms can serve millions of students simultaneously with consistent quality.

1.3 MARKET DYNAMICS

The Indian EdTech market is shaped by several forces. Rising smartphone penetration, affordable internet (Jio revolution), the post-COVID shift to online education, and increasing aspiration among Indian youth for IITs, NITs, and medical colleges are key demand drivers. The JEE/NEET preparation segment alone represents millions of students annually. Competition is intense, with players like BYJU'S, Unacademy, Vedantu, and Allen competing alongside PW. However, PW has differentiated itself through price leadership - delivering world-class content at prices accessible to every Indian student.

1.4 INDUSTRY CHALLENGES

The EdTech industry faces several challenges. Student retention and completion rates remain low for self-paced content. Building trust in online degrees and certifications is an ongoing challenge. Infrastructure limitations in rural areas restrict access. Additionally, balancing quality with scale - ensuring every student gets personalized attention - remains a complex operational and pedagogical problem. Regulatory scrutiny around admission processes and fee structures has also increased.

1.5 FUTURE PROSPECTS

The future of Indian EdTech is promising. Hybrid learning models combining online and offline education (like PW's own brick-and-mortar centres) are gaining momentum. AI-powered tutoring, vernacular content in regional languages, skill-based learning for employment, and international expansion are emerging frontiers. PW's PWIOI initiative - building a full-fledged innovation institute - represents the next evolution of EdTech from pure content delivery to institution building.

2. EDTECH INDUSTRY ANALYSIS

2.1 GLOBAL EDTECH INDUSTRY ANALYSIS

2.1.1 Market Size and Growth

The global EdTech market was valued at approximately USD 254 billion in 2023 and is projected to grow at a CAGR of 13.6%, reaching over USD 600 billion by 2030. Growth is driven by increasing internet penetration, government digitization programs, and rising demand for upskilling and reskilling in the post-pandemic world.

2.1.2 Key Drivers of Global EdTech

• Digital Infrastructure: Expanding broadband and mobile internet access globally enables online learning at scale.

• COVID-19 Acceleration: The pandemic forced a rapid shift to digital learning, permanently changing user behavior.

• AI and Personalization: Adaptive learning algorithms improve outcomes by tailoring content to individual student needs.

• Corporate Upskilling: Enterprises increasingly invest in EdTech platforms for employee training and development.

• Government Initiatives: Programs like India's NEP 2020 and Digital India promote technology-enabled education.

2.1.3 Key Global Players

• Coursera: A leading MOOC platform partnering with top global universities for online degrees and certificates.

• Duolingo: The world's most downloaded language learning app, known for gamification and AI-driven personalization.

• Udemy: A marketplace for professional skills courses with over 200,000 courses across domains.

• Khan Academy: A non-profit offering free world-class education to anyone, anywhere.

• PhysicsWallah (PW): India's fastest-growing EdTech unicorn, focused on affordable JEE/NEET preparation.

2.2 INDIAN EDTECH INDUSTRY ANALYSIS

2.2.1 Market Size and Growth

India's EdTech sector is one of the world's fastest growing, with the market expected to reach $30 billion by 2030. India has over 500 EdTech startups, and the K-12 and test preparation segments are the largest verticals. PW alone crossed a $2.8 billion valuation within years of expanding beyond YouTube, reflecting the enormous demand for affordable quality education.

2.2.2 Key Indian Players

a) PhysicsWallah (PW): India's most affordable and fastest-growing EdTech platform, valued at $2.8B+, known for JEE and NEET preparation.

b) BYJU'S: India's largest EdTech company by historical valuation, offering K-12 and competitive exam solutions.

c) Unacademy: A live-learning platform with thousands of educators and subscription-based learning.

d) Vedantu: A live tutoring platform focused on interactive doubt-solving and small-batch learning.

e) Allen Career Institute: The dominant offline coaching brand expanding aggressively into EdTech.

3. OVERVIEW OF PHYSICSWALLAH (PW)

3.1 HISTORY AND BACKGROUND

PhysicsWallah (PW) was founded by Alakh Pandey in 2014 as a YouTube channel dedicated to teaching Physics for JEE and NEET aspirants. What started as free video lectures quickly grew into one of India's most beloved educational brands, known for Alakh Sir's engaging teaching style, affordable fees, and genuine care for students' success. In 2020, PW launched its app and shifted to a subscription-based model while keeping prices at a fraction of traditional coaching. In 2022, PW became India's 101st unicorn after raising $100 million at a $2.8 billion valuation - making it one of the fastest Indian startups to achieve unicorn status.

3.2 KEY HISTORICAL MILESTONES

• 2014: Alakh Pandey starts PhysicsWallah YouTube channel for free JEE/NEET Physics content.

• 2018: Channel reaches 1 million subscribers; becomes India's most popular free Physics channel.

• 2020: PW App launched; subscription-based courses offered at INR 999-5,000 (vs INR 1-2 lakh for offline coaching).

• 2021: Expanded beyond Physics to cover all JEE/NEET subjects with 100+ educators.

• 2022: Raised $100M at $2.8B valuation - India's 101st unicorn. Prateek Maheshwari joins as co-founder.

• 2023: Launched PW Vidyapeeth and PWIOI (PhysicsWallah Institute of Innovation); expanded to offline centres.

• 2024-2026: PWIOI Phase 1 campus development; IOI hybrid model discussions with leadership.

3.3 VISION AND VALUES

3.3.1 Vision

PhysicsWallah aims to be 'The Education of India' - making world-class education accessible and affordable to every Indian student, regardless of geography or economic background. Their vision is to create a generation of problem-solvers, innovators, and leaders through technology-enabled, student-first education.

3.3.2 Core Values

• Student First: Every decision is made with the student's learning outcomes in mind.

• Affordability: Quality education should not be a privilege; it must be accessible to all.

• Innovation: Continuously experimenting with new teaching methods, technology, and formats.

• Authenticity: Real teachers, real care, real results - no marketing gimmicks.

• Scale with Quality: Growing rapidly without compromising the quality of learning.

3.4 KEY PROGRAMS AND INITIATIVES

a) PW App (JEE/NEET Preparation): The core product - video lectures, practice questions, live sessions, and mock tests for competitive exam aspirants.

b) PWIOI (PhysicsWallah Institute of Innovation): A full-fledged innovation institute being built to give India world-class engineering education at PW's signature affordable rates.

c) PW Vidyapeeth: An initiative for school-level education (Class 6-12), targeting foundational learning.

d) PW Skills: Upskilling programs for working professionals in Data Science, AI, and emerging technologies.

e) PW Centers (Offline): Physical coaching centres across India bringing PW's affordable model to brick-and-mortar locations.

4. THE PWIOI TEAM - PRODUCT & OPERATIONS

4.1 INTRODUCTION TO PWIOI

PWIOI - PhysicsWallah Institute of Innovation - is PW's flagship initiative to build a world-class educational institution that combines academic rigor with innovation and research. The PWIOI Product and Operations team is responsible for managing the entire operational backbone of PWIOI: from student admissions and sales analytics to website management and automation of internal processes.

4.2 ROLE AND RESPONSIBILITIES OF THE PRODUCT TEAM

The PWIOI Product Engineering team operates at the intersection of data, technology, and operations. The team's primary responsibilities include:

4.2.1 Business Intelligence and Dashboard Management

Building, maintaining, and iterating on Looker Studio dashboards that track admissions, sales funnels, walk-in data, registration counts, and LY/CY comparisons. These dashboards are used by senior leadership including Neeraj Sir, Samreen Mam, Davinder Sir, and Alakh Sir's team for strategic decision-making.

4.2.2 Workflow Automation

Developing Python scripts and Google Sheets automations to replace manual, repetitive data tasks - including student score mapping, BTL/Mitra tracking, scholarship status checking, ERP synchronization, and report generation. Automation directly reduces man-hours spent on non-value-adding activities.

4.2.3 Marketing Automation and Community Management

Building and maintaining the PWIOI Telegram bot for channel promotions, user notifications, and lead engagement. The team also connects directly with prospective students through Telegram calls and messaging to support admission conversions.

4.2.4 Website Content Management (GCMS)

Managing content updates on the PWIOI website (pwioi.com) through the GCMS platform - including teacher profiles, campus photos, curriculum pages, SOT page updates, shorts, fee structures, and home page widgets.

4.2.5 Product Documentation (BRDs)

Writing Business Requirement Documents for new product features and modules, including the Loan Module, Bonafide Module, and Refund Page. BRDs define the functional requirements, user flows, and business logic for engineering teams to build features.

4.3 TECHNOLOGY STACK AND TOOLS

Category	Tools / Technologies
BI & Dashboards	Looker Studio (Google Data Studio)
Automation	Python, Google Apps Script, Streamlit
Data / Sheets	Google Sheets (XLOOKUP, ImportRange, ArrayFormula)
Bot Development	Telegram Bot API, Python
Website CMS	GCMS (Internal CMS), pwioi.com
Video Processing	Streamlit, Python (PDF parsing)
Reporting	Gmail, Google Sheets, PDF Reports
Communication	Telegram, Email, Scrum Meetings, WBR

4.4 ORGANIZATIONAL STRUCTURE

The PWIOI team is structured around functional leads who oversee different verticals. The intern (AI Product Engineer) works directly under senior managers and collaborates cross-functionally with data, marketing, admissions, and tech teams:

- Neeraj Sir - Sales, IOI Summary, Walk-in Reporting, Funnel Analytics
- Davinder Sir - Admission Samadhan, BTL/Mitra Tracking, Daily Operations
- Abhishek Sir - Excel Automations, Score Mapping, Dashboard Alignment
- Shimran Mam - Mettle Automation, Sheet Workflows, Scholarship Tracking
- Samreen Mam - Sales Dashboard, Walk-in Reports, Dashboard Reviews
- Khushi Mam - Mitra Dashboard, Student Tracking
- Amrit Sir - Website (GCMS) Content Updates

5. INTERNSHIP EXPERIENCE AT PHYSICSWALLAH

My internship at PhysicsWallah (PW) as a AI Product Engineer in the PWIOI team has been a transformative and enriching experience. Over the course of the internship, I engaged in diverse activities spanning Business Intelligence, Automation Engineering, Website Management, and Product Documentation - contributing to real operational outcomes used by the leadership team.

5.1 OBJECTIVE

To contribute to PhysicsWallah's PWIOI product and operations team by building data-driven dashboards, automating repetitive workflows, managing digital assets, and supporting product documentation - while developing hands-on experience in an EdTech product environment.

5.2 DESCRIPTION OF DUTIES

During my internship, I worked at PW's Noida office and collaborated remotely with multiple team members. My duties spanned four major categories:

5.2.1. Daily Walk-in Report Mailing

Sent daily walk-in reports to all senior ADs and the leadership team via email. This involved compiling data from the admission tracking sheets, verifying numbers, and dispatching the standardized report each morning. This was a critical daily ritual ensuring leadership had current admission data to make decisions.

5.2.2. Looker Studio Dashboard Development

Designed and deployed multiple end-to-end dashboards on Google Looker Studio covering CEE/FTP admission funnels, walk-in data, LY vs CY comparisons, city-wise analysis, sales funnel, registration counts, and targeted vs achieved metrics. Each dashboard involved data source setup, calculated fields, custom filters, and layout design as per stakeholder requirements.

5.2.3. Excel and Google Sheets Automation

Built Python and Google Apps Script automations for: student academic score mapping (10th/12th/JEE), BTL+Mitra tracking with ME-wise date filters, scholarship status checker, ERP synchronization, counselling preference column updates, and summary report generation. These automations eliminated hours of manual data entry across multiple teams.

5.2.4. Mettle Video Downloader Automation

Developed a Streamlit-based web tool (deployed live) to extract candidate assessment videos from Mettle exam PDFs. The tool parses PDFs, extracts video links, downloads files with email-based naming conventions, and organizes output for the hiring team. Used to evaluate 200+ candidate videos during the internship.

5.2.5. Telegram Bot - PWIOI Channel Automation

Built and maintained a custom Telegram bot for the PWIOI marketing channel. The bot automates promotional messages and link sharing when new members join the channel. Extended functionality to handle all message formats, link types, and notification schedules as required by Neeraj Sir. Token was also updated and bot was redeployed mid-internship.

5.2.6. GCMS Website Management (pwioi.com)

Managed content updates on the PWIOI official website using the GCMS platform: added Microsoft and Apple widgets to the home page, uploaded campus and cultural photos, updated the SOT (School of Technology) page with teacher profiles, curriculum, fee structure, and shorts/videos. Sent update confirmation emails to all relevant ADs after each deployment.

5.2.7. BRD Writing - Product Documentation

Authored Business Requirement Documents for three product modules: Loan Module, Bonafide Module, and Refund Page. BRDs included user stories, functional requirements, edge cases, and process flows. Participated in review sessions with Abhishek Sir and Malayaj Sir to refine requirements.

5.2.8. Data Integrity and Sheet Fixes

Replaced ImportRange formulas with XLOOKUP in CEE and FTP sheets to eliminate NA errors. Corrected calculated fields, filters, and formula logic across multiple dashboards and sheets. Verified dashboard data against source sheets during WBR preparation in collaboration with Shimran Mam.

5.2.9. CEE File Video Analysis

Processed and evaluated 200+ CEE exam PDFs to check video availability. Generated summary reports of video availability status across all CEE files. Automated video extraction and FTP file handling as part of the CEE assessment pipeline.

5.2.10. Telegram Lead Calling

Connected with 10+ prospective PWIOI students through direct Telegram calls. Explained the program, addressed queries, and supported the admissions pipeline as part of the community engagement effort.

5.2.11. JEE Rank Predictor Data Pipeline

Dumped overall JEE rank predictor data from the platform, cleaned the dataset, and prepared it for drip campaign use. Data cleaning involved handling missing values, standardizing formats, and removing duplicates to ensure data quality for marketing automation.

5.2.12. KT Transfer

Conducted a structured Knowledge Transfer (KT) session for dashboard handover to Sarlean, covering all active dashboards, their data sources, refresh schedules, and filter logic. Ensured complete documentation for continuity.

5.3 WEEK-WISE TASK SUMMARY

Period	***Key Activities***
Week 1	HR onboarding, PW work culture orientation, credentials setup (mail, Darwinbox), offer letter acceptance, introduction to Abhishek Sir and Neeraj Sir.
Week 2-3	First Telegram bot v1 built, Excel automations for Shimran Mam and Abhishek Sir (score mapping), Mettle video downloader tool development.
Week 4-6	MOU Dashboard (Looker), Admission Samadhan Dashboard for Davinder Sir, Sales Dashboard Phase and IOI Summary Dashboard for Neeraj Sir.
Week 7-10	City-wise funnel dashboard, daily walk-in report mailing started, CEE file automation, GCMS website updates (SOT page, teachers, curriculum).
Week 11-14	BRDs for Loan, Bonafide, Refund modules; BTL+Mitra dashboard with 3 new columns; JEE rank predictor data pipeline; XLOOKUP fixes in CEE/FTP sheets.
Week 15+	CAP Dashboard, LY/CY monthly comparison, Refer & Earn funnel dashboard, Adda49 dashboard, BTL/Mitra automation, KT transfer to Sarlean, WBR preparation.

5.4 KEY PROJECTS HANDLED

5.4.1. PWIOI Admission & Sales Dashboards

Built and maintained 15+ dashboards on Google Looker Studio for the PWIOI leadership team. Key dashboards include the Admission Samadhan Dashboard (walk-in and admission tracking), CEE/FTP Funnel Dashboard (registration to admission funnel analysis), PWIOI 2026 Sales Dashboard (sales funnel with city-wise breakdown), LY/CY Comparison Dashboard (year-over-year analysis), and the JM1 Dashboard. Each dashboard involved custom data connections, calculated metrics, multi-level filters, and layout designed for executive viewing.

Admission Samadhan: https://datastudio.google.com/reporting/98d718ae-8282-4013-be53-828fd9e968e5

CEE & FTP Funnel: https://datastudio.google.com/reporting/1d8f4a35-c98b-4670-952d-6cb70f3ccb62

JM1 Dashboard: https://datastudio.google.com/reporting/ff2d5629-100c-4215-91b1-02847eeefc6e

BTL+Mitra: https://datastudio.google.com/reporting/374c5ac1-c15a-476a-b963-1153a9a4bcf3

5.4.2. Mettle Video Downloader Automation

Developed a production-grade Streamlit web application to automate video extraction from Mettle exam PDFs. The tool parses PDF documents to extract embedded video links, downloads the videos, applies email-based naming conventions for organization, and handles FTP file management. This tool processed 200+ candidate assessment videos, significantly improving the recruitment pipeline efficiency for the PWIOI hiring team.

Live Tool:https://mettleautomation.streamlit.app/

5.4.3. Telegram Bot - PWIOI Marketing Automation

Designed and deployed a custom Telegram bot for the PWIOI marketing channel. The bot automates promotional content delivery, sends notifications to new channel members, handles link sharing, and supports all message formats required by the marketing team. The bot was iterated multiple times based on feedback and was redeployed with updated tokens mid-internship. Connected 10+ leads through direct Telegram calling.

5.4.4. Excel and Google Sheets Automations (7+)

Delivered 7+ Python and Sheets automation solutions: (1) Student score mapping automation for 10th, 12th, and competitive exam scores using ID-based matching; (2) BTL+Mitra tracking automation with date-filter and ME-wise breakdown; (3) Scholarship status checker; (4) ERP synchronization script; (5) Admission summary formula updates; (6) Davinder Sir's summary report automation; (7) Counselling preference column automation.

5.4.5. Additional Dashboards

Built multiple additional dashboards: MOU Dashboard (Looker Studio), BTL & Mitra Tracking Dashboard, College Listing Tracker, CAP Dashboard, Refer & Earn Funnel Dashboard, Review Adda49 Dashboard (coupon code analysis), Mitra Dashboard for Khushi Mam, and Samreen Mam's Sales Dashboard.

College Listing: https://datastudio.google.com/reporting/cb65eb49-3e9c-41ba-a3e4-cb2a8837b35f

Review Adda49: https://datastudio.google.com/reporting/14e0832d-6c51-4eaa-99d4-029e331b29a8

Priyanka Mam CEE+FTP: https://datastudio.google.com/reporting/54af00e9-6dee-4cd1-8924-e93bb8585be6

5.4.6. PWIOI Website Management

Managed regular content updates on the official PWIOI website (pwioi.com) via the GCMS platform. Updates included: adding Microsoft and Apple technology partner widgets to the home page, uploading campus and cultural event photos, building out the SOT (School of Technology) page with complete curriculum, teacher profiles, fee structure, and short-form videos. Sent deployment confirmation emails to all ADs after each update cycle.

5.4.7. BRD Documentation

Authored three Business Requirement Documents (BRDs) for upcoming product modules: (1) Loan Module - for student loan facilitation integrated with the admission portal; (2) Bonafide Module - for generating bonafide certificates for enrolled students; (3) Refund Page - defining the refund initiation and processing workflow. Each BRD was reviewed and iterated with Abhishek Sir and Malayaj Sir.

5.5 LEARNING OUTCOMES

The internship provided invaluable learning experiences across technical, analytical, and professional domains:

5.5.1. Real-world Business Intelligence

Learned to build dashboards that leadership actually uses - not just technically sound but stakeholder-driven with the right metrics, layouts, and filters. Understood how to translate business questions ('show me city-wise funnel this week') into data models and visual outputs.

5.5.2. Automation as a Product Mindset

Writing Python scripts to replace manual work taught me to think about automation as a product - with error handling, naming conventions, reusability, and documentation. Good automation is invisible to the end user.

5.5.3. Cross-functional Collaboration

Working with 8+ stakeholders (Abhishek Sir, Neeraj Sir, Davinder Sir, Shimran Mam, Samreen Mam, Khushi Mam, Priyanka Mam, and Mehul Sir) taught me to manage competing priorities, communicate requirements clearly, and iterate quickly in a fast-paced EdTech environment.

5.5.4. Product Thinking

Writing BRDs for Loan, Bonafide, and Refund modules gave hands-on exposure to how product features are scoped, documented, and aligned with engineering teams. Thinking from the user's perspective while defining functional requirements is a skill I deepened during this experience.

5.5.5. Operational Efficiency

Understood the significance of operational efficiency in a high-growth EdTech environment - where decisions need to be made daily based on data. The role of BI and automation in enabling this speed became very clear.

In conclusion, my PhysicsWallah internship was transformative - equipping me with real-world skills in data engineering, BI, and product operations. Working alongside experienced professionals on high-impact projects has been immensely rewarding. The insights and professional relationships developed will undoubtedly shape my career in AI, data, and EdTech product roles.

6. SUGGESTIONS FOR IMPROVEMENT

The PWIOI Product team has established strong operational foundations. However, based on observations during the internship, the following areas present significant opportunities for further improvement in efficiency, data quality, and decision-making speed.

6.1 CENTRALIZED DASHBOARD GOVERNANCE

6.1.1 Current Situation

Dashboard ownership is distributed across multiple stakeholders without a centralized governance framework. This leads to duplication - multiple dashboards tracking the same metrics with different numbers due to varying data sources or filter logic.

6.1.2 Proposed Improvement

Establish a 'Single Source of Truth' dashboard architecture with one master data source per metric domain (admissions, sales, walk-ins). All dashboards should pull from the same cleaned, validated dataset. A Dashboard Registry document should be maintained listing all active dashboards, their owners, data sources, and refresh schedules.

6.1.3 Benefits

• Eliminates data discrepancies between dashboards.

• Reduces time spent debugging conflicting numbers in WBR meetings.

• Enables faster onboarding of new team members to the data infrastructure.

6.2 AUTOMATED DAILY REPORT GENERATION

6.2.1 Current Situation

Daily walk-in reports and city-wise summary reports are compiled manually from sheets and sent via email each morning. This is time-consuming and prone to delays if the responsible person is unavailable.

6.2.2 Proposed Improvement

Build an automated reporting pipeline using Google Apps Script or Python + Gmail API that pulls data from the master sheet at a scheduled time each morning, generates a formatted email report, and auto-dispatches it to the relevant distribution list - with zero manual intervention required.

6.2.3 Benefits

• Eliminates daily manual effort for report preparation.

• Ensures consistent, on-time delivery regardless of individual availability.

• Allows the team to focus on analysis rather than report formatting.

6.3 ADMISSION LEAD SCORING SYSTEM

6.3.1 Current Situation

Telegram leads and walk-in enquiries are followed up manually without any systematic scoring or prioritization. High-intent leads receive the same treatment as cold contacts.

6.3.2 Proposed Improvement

Implement a simple lead scoring model based on engagement signals (Telegram activity, walk-in visits, test scores, program queries) to prioritize follow-up. A scored lead list could be auto-generated weekly and shared with the admissions team.

6.3.3 Benefits

• Higher conversion rates by focusing effort on high-intent leads.

• Data-driven admission follow-up replacing gut-feel prioritization.

• Measurable funnel improvement over time.

6.4 AUTOMATED DATA VALIDATION LAYER

6.4.1 Current Situation

Dashboard data occasionally has errors - NA values, incorrect mappings, or stale data - that are only caught during manual review or WBR meetings. Multiple correction cycles waste hours before each major meeting.

6.4.2 Proposed Improvement

Build an automated data validation script that runs daily before reports are dispatched. It should flag NA values, outliers, mismatched IDs, and formula errors - and notify the responsible team member for correction before data reaches stakeholders.

6.4.3 Benefits

• Prevents embarrassing data errors in leadership reviews.

• Reduces time spent on manual data verification before WBR.

• Builds confidence in the data across the team.

In conclusion, the recommendations in this section aim to enhance the PWIOI Product team's operational efficiency through smarter automation, better data governance, and AI-assisted lead management. Implementing these suggestions will improve decision-making speed, reduce manual workload, and strengthen PW's data-driven culture.

7. CONCLUSION

The internship at PhysicsWallah (PW) in the PWIOI Product Engineering team provided an exceptional opportunity to gain practical experience at the intersection of data engineering, business intelligence, and EdTech product operations. This experience offered a deep understanding of how one of India's leading EdTech unicorns operates internally - from data pipelines and dashboards to automation scripts and product documentation.

The analysis of India's EdTech industry revealed its dynamic and rapidly evolving nature, driven by massive student aspirations, digital penetration, and the affordability revolution led by companies like PhysicsWallah. PW's PWIOI initiative stands as a bold step toward institution building - a bridge between EdTech and traditional higher education.

Key takeaways from the internship include the critical importance of data-driven decision-making in a high-growth EdTech environment, the power of automation in eliminating operational bottlenecks, and the role of cross-functional collaboration in delivering results at scale. Every dashboard, script, and BRD I built was used by real stakeholders to make real decisions - which made the work deeply rewarding.

During my time at PW, I actively built dashboards used weekly by senior leadership, automated workflows that saved hours of manual effort, developed a live production tool for video extraction, and authored product documents that will inform future engineering work. This hands-on experience gave me a comprehensive understanding of what product engineering looks like in a fast-paced startup environment.

The strategic recommendations provided in this report - centralized dashboard governance, automated reporting pipelines, lead scoring, and data validation - aim to further strengthen PW's operational infrastructure. These initiatives will improve efficiency and ensure PW's data systems scale as gracefully as its student base does.

Overall, this internship has been instrumental in bridging the gap between academic learning in Data Science and AI at IIT Guwahati and real-world product engineering. The skills developed, the relationships built, and the impact delivered will be invaluable as I pursue a career in AI, data engineering, and product development.

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